Quantification: 699_HYP_HP_z20c1-2
==================================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	11
interior lacuna count	11
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 11)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	1438.44	1239	799.765	px^2
major_axis_length_px	lacuna shape	70.1037	63.7	22.8233	px
minor_axis_length_px	lacuna shape	33.4837	30.7419	15.9453	px
aspect_ratio	lacuna shape	2.5837	1.665	1.5667	unitless
eccentricity	lacuna shape	0.8208	0.7995	0.1366	unitless
solidity	lacuna shape	0.7565	0.7599	0.117	unitless
roots_count	headline	11	8	13.8636	count
ring_length_r30_px	headline	392.091	343	417.233	px
ring_length_r60_px	reported	765.727	741	639.976	px
owned_length_px	ownership-dependent	816.332	774.713	904.822	px
edge_count	ownership-dependent	46.5455	44	54.4213	count
mean_edge_length_px	ownership-dependent	19.5203	18.4615	6.8357	px
perimeter_px	normalised	328.647	185.581	382.104	px
ring_area_r30_px2	normalised	9852.09	7465	6872.98	px^2
ring_area_r60_px2	normalised	22578.3	19284	11474.9	px^2
in_frame_fraction_r30	normalised	0.9784	1	0.0715	unitless
in_frame_fraction_r60	normalised	0.9565	1	0.1056	unitless
ring_density_r30	normalised	0.0360491	0.0331272	0.0172511	px^-1
ring_density_r60	normalised	0.0316939	0.0332926	0.0138651	px^-1
roots_per_100px_perimeter	normalised	3.2694	3.5528	1.4521	per 100 px
ring_attached_length_r30_px	network v2	347.727	216	412.7	px
ring_attached_length_r60_px	network v2	591.455	352	644.197	px
ring_length_w_r30_px	network v2	438.975	386.433	470.169	px
ring_length_w_r60_px	network v2	860.389	827.784	728.157	px
sholl_crossings_r10	network v2	8.7273	6	10.1498	count
sholl_crossings_r20	network v2	8.9091	8	8.1542	count
sholl_crossings_r30	network v2	11	10	8.8204	count
loop_count	loops, report only	1.8182	1	2.892	count
loop_edge_share	loops, report only	0.099	0.0682	0.1032	share
ring_width_mean_r30_px	width	7.3278	7.0585	1.5525	px

Per field
----

measure	value	units
analysed_area_px2	1.02596e+06	px^2
total_skeleton_length_px	18204	px
canalicular_length_density_per_px	0.0177434	px^-1
junction_count	339	count
junction_density_per_px2	0.00033042	px^-2
lacuna_count	11	count

lacunae_per_px2	1.072e-05	px^-2
skeleton_component_count	339	count
mean_component_length_px	53.6991	px
median_component_length_px	19	px
field_length_density_w_per_px	0.0198989	px^-1
field_density_without_flagged_per_px	0.0167541	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	964	px
unattached_component_count	48	count
canalicular_length_density_attached_per_px	0.0177434	px^-1
field_length_density_with_unattached_per_px	0.018683	px^-1
loop_count	24	count
loop_edge_share	0.0884	share
edge_count	1018	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.9355	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	13941	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1079	46.6189	34.6236	1.3464	0.6696	0.7599	-0.6851	205.172	387.171	False
2	3222	86.3749	58.8691	1.4672	0.7318	0.7263	-0.9453	298.969	636.32	False
3	1329	56.9074	38.8202	1.4659	0.7312	0.7408	-0.6935	386.281	385.088	False
4	574	63.7	12.3761	5.147	0.9809	0.8658	-1.4148	410.084	638.57	False
5	961	72.5694	18.5814	3.9055	0.9667	0.8076	1.4502	539.664	699.95	False
6	1511	109.373	23.0638	4.7422	0.9775	0.6218	1.4725	559.445	447.167	False
7	898	39.9158	30.7419	1.2984	0.6378	0.8635	1.334	631.296	609.345	False
8	2133	94.4671	56.7385	1.665	0.7995	0.538	0.9278	716.599	755.109	False
9	1239	61.0069	27.5385	2.2153	0.8923	0.8844	-1.1926	1001.21	883.603	False
10								41.4048	620.589	False
11								283.342	208.259	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	241	741	991.553	51	19.4422	8.4329
2	8	343	755	809.838	50	16.1968	6.2878
3	9	401	809	774.713	44	17.6071	7.0585
4	6	192	446	345.238	17	20.3081	6.6589
5	4	370	654	443.563	29	15.2953	7.2478
6	16	545	972	1163.08	63	18.4615	7.842
7	7	348	808	1017.65	49	20.7683	6.0891
8	9	161	333	127.669	10	12.7669	6.6426

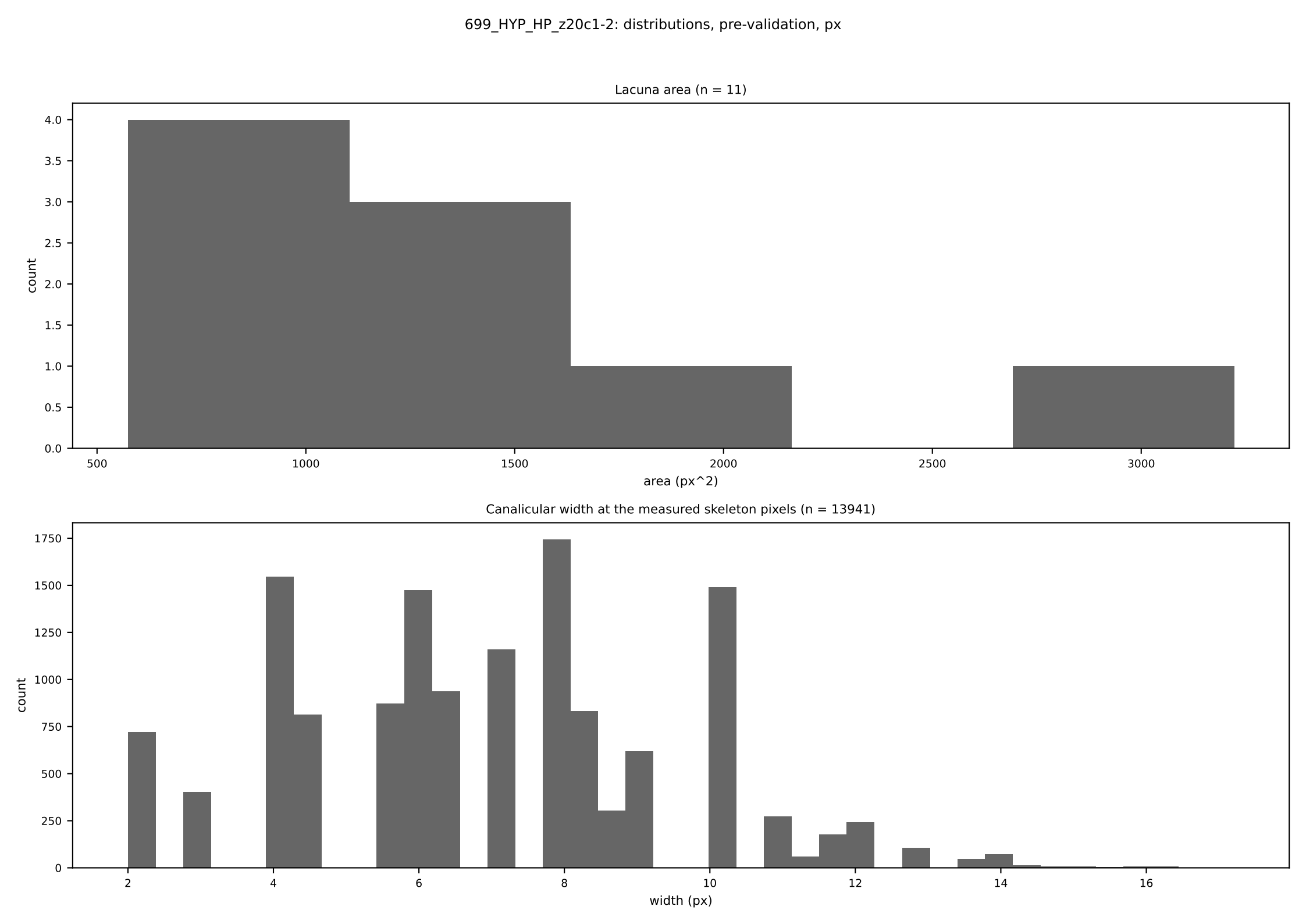
9	2	35	62	18.4853	1	18.4853	5.146
10	1	107	328	38.8995	1	38.8995	11
11	51	1570	2515	3248.96	197	16.4922	8.2003

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	185.581	7275	19284	1	1	0.0331272	0.0384256	4.3108
2	418.517	11182	24664	1	1	0.0306743	0.0306114	1.9115
3	221.179	7585	17507	1	1	0.0528675	0.0462101	4.0691
4	140.326	6761	16630	1	1	0.0283982	0.026819	4.2758
5	165.74	7465	19644	1	1	0.0495646	0.0332926	2.4134
6	316.149	10460	25779	1	1	0.0521032	0.0377051	5.0609
7	132.267	6479	17177	1	1	0.053712	0.0470397	5.2923
8	363.747	11289	26942	1	1	0.0142617	0.0123599	2.4742
9	163.681	5591	13507	0.7628	0.6662	0.00626006	0.00459021	1.2219
10	72.4264	4788	13028	1	0.8554	0.0223475	0.0251765	1.3807
11	1435.5	29498	54199	1	1	0.0532239	0.0464031	3.5528

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	203	646	268.966	827.784	4	6	10
2	216	332	386.433	843.11	9	9	12
3	379	668	448.363	905.997	7	10	9
4	184	253	212.297	498.818	3	6	4
5	255	352	403.936	727.014	7	8	12
6	543	767	624.014	1112.56	15	13	12
7	298	684	387.25	905.169	6	12	13
8	134	204	175.125	362.25	5	2	6
9	24	24	40.4558	70.8406	0	0	1
10	74	188	117.255	359.764	3	2	7
11	1515	2388	1764.63	2850.97	37	30	35


699_HYP_HP_z20c1-2: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 13941)
count
width (px)